SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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CSA37- SOFTWARE TESTING

EXP:5

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Requirement Analysis and Negative Test Scenarios and Test Cases for an E-Commerce Application

Aim

To study the requirements of an E-Commerce Application, and to prepare negative test scenarios and test cases that confirm the application handles invalid inputs and wrong user actions safely.

Problem Statement

An e-commerce application lets customers browse products, add them to a cart, pay online and track their orders. It handles personal data, payments and stock, so a defect can cause lost sales, wrong charges or loss of customer trust.

Testing is needed to check that the application:

- Customers can register, log in, search for products and place orders correctly.
- Invalid actions are rejected safely with clear messages.
- Cart, payment and order handling follow the stated requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	A user can register with name, email, mobile number and password. The email must be unique.
FR2	A registered user can log in with valid credentials and log out.
FR3	The user can search products by name or category and get relevant results.
FR4	The user can filter and sort results by price, brand and rating.
FR5	The product page shows images, price, description, rating and stock availability.
FR6	The user can add a product to the cart, change its quantity and remove it. Out-of-stock items cannot be added.
FR7	The cart shows the correct item prices and total. A valid coupon reduces the total correctly.
FR8	Checkout needs a valid delivery address. The user can save more than one address.
FR9	Payment is possible by card, UPI, net banking or cash on delivery.
FR10	After a successful payment, the system shows an order ID and sends an email or SMS confirmation.
FR11	The user can track the order status and view the order history.
FR12	An order can be cancelled before it is shipped. The refund goes to the original payment method.

2 Software and environment requirements

Item	Details
Application	Web application (React front end, Java/Node.js back end) and mobile browser view
Database	MySQL / MongoDB (users, products, carts, orders)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, Postman, bug tracker (Jira)
Test data	Sample users, products with and without stock, coupon codes, sandbox card and UPI details

Negative Test Scenarios

A negative test scenario checks that the system rejects invalid input or a wrong action, shows a clear message and stays stable and secure.

Scenario ID	Module	Test Scenario	Requirement
EC-NS01	Registration	Register with an email that already exists	FR1
EC-NS02	Registration	Register with an invalid email and weak password	FR1
EC-NS03	Registration	Submit the form with mandatory fields blank	FR1
EC-NS04	Login	Log in with a wrong password	FR2
EC-NS05	Login	Log in with an unregistered email	FR2
EC-NS06	Search	Search for a product that does not exist	FR3
EC-NS07	Search	Search with special characters and a script string	FR3
EC-NS08	Cart	Add an out-of-stock product to the cart	FR6
EC-NS09	Cart	Set an invalid quantity	FR6
EC-NS10	Cart	Apply an invalid or expired coupon	FR7
EC-NS11	Checkout	Check out with an empty cart or no address	FR8
EC-NS12	Payment	Pay with an invalid or expired card	FR9
EC-NS13	Payment	Payment fails or is cancelled midway	FR9, FR10
EC-NS14	Cancellation	Cancel an order after it has been shipped	FR12

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
EC-NT01	EC-NS01	1. Open the registration page 2. Enter an already registered email with other valid details 3. Click Register	ravi.k@example.com	Registration is rejected with “Email already registered”. No duplicate account is created.
EC-NT02	EC-NS02	1. Enter an email without @ 2. Enter a 3-character password 3. Click Register	ravi.example.com; 123	Field errors for the invalid email and weak password are shown. Account is not created.
EC-NT03	EC-NS03	1. Leave all fields empty 2. Click Register	Blank	Mandatory-field messages are displayed and the form is not submitted.
EC-NT04	EC-NS04	1. Enter a registered email and a wrong password 2. Click Login	ravi.k@example.com; Wrong@123	Login fails with “Invalid email or password”. The user is not logged in.
EC-NT05	EC-NS05	1. Enter an email that is not registered 2. Enter any password 3. Click Login	unknown@example.com; Test@123	Login fails with a clear error message and no account details are revealed.
EC-NT06	EC-NS06	1. Type a meaningless product name 2. Press Enter	xyzqwerty123	“No results found” is shown with suggestions. The page does not crash.
EC-NT07	EC-NS07	1. Type special characters in the search box 2. Press Enter	<script>alert(1)</script>; ' OR 1=1 --	The input is treated as plain text. No script runs and no data is exposed.
EC-NT08	EC-NS08	1. Open an out-of-stock product 2. Try to click Add to Cart	Out-of-stock product	Add to Cart is disabled or a “Currently unavailable” message is shown.
EC-NT09	EC-NS09	1. Open the cart 2. Enter quantity 0, then -1, then a value above stock	0; -1; 9999	The quantity is rejected, and the maximum available stock is shown.
EC-NT10	EC-NS10	1. Add items to the cart 2. Enter a wrong or expired coupon 3. Click Apply	SAVE99; EXPIRED20	“Invalid or expired coupon” is shown and the total does not change.
EC-NT11	EC-NS11	1. Open checkout with an empty cart 2. Then proceed with items but no address	Empty cart; blank address	Checkout is blocked, and the user is asked to add items or an address.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
EC-NT12	EC-NS12	1. Select Card 2. Enter an expired card or a wrong CVV 3. Click Pay	4111 1111 1111 1111; expired 01/22	Payment is declined with a clear message and the order is not placed.
EC-NT13	EC-NS13	1. Start a UPI payment 2. Reject or cancel it in the sandbox	failure@upi	Order is not confirmed, the user is not charged and can retry. No duplicate charge occurs.
EC-NT14	EC-NS14	1. Open an order with status Shipped 2. Click Cancel	Shipped order	Cancel is disabled, or the user is told to request a return after delivery.

Result / Conclusion

Result: 14 negative test scenarios and 14 test cases were prepared for the E-Commerce Application, covering 9 of the 12 functional requirements (no negative scenario for FR4, FR5, FR11). The test cases by module are:

Module	Test Scenarios	Test Cases
Registration	3	3
Login	2	2
Search	2	2
Cart	3	3
Checkout	1	1
Payment	2	2
Cancellation	1	1
Total	14	14

Conclusion: The negative test cases cover invalid inputs, boundary violations, failed payments or operations, and unauthorized actions in the E-Commerce Application. When they are executed on a build, they confirm that the system rejects wrong input safely, shows clear messages, and keeps the data and the user’s session secure. Together with the positive test cases, they give a complete set of tests for the stated requirements.